

I used my own sales_clean.csv from module 3.

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("sales_clean.csv", dtype={"zip": str})
print(df.shape)
```

(292, 6)

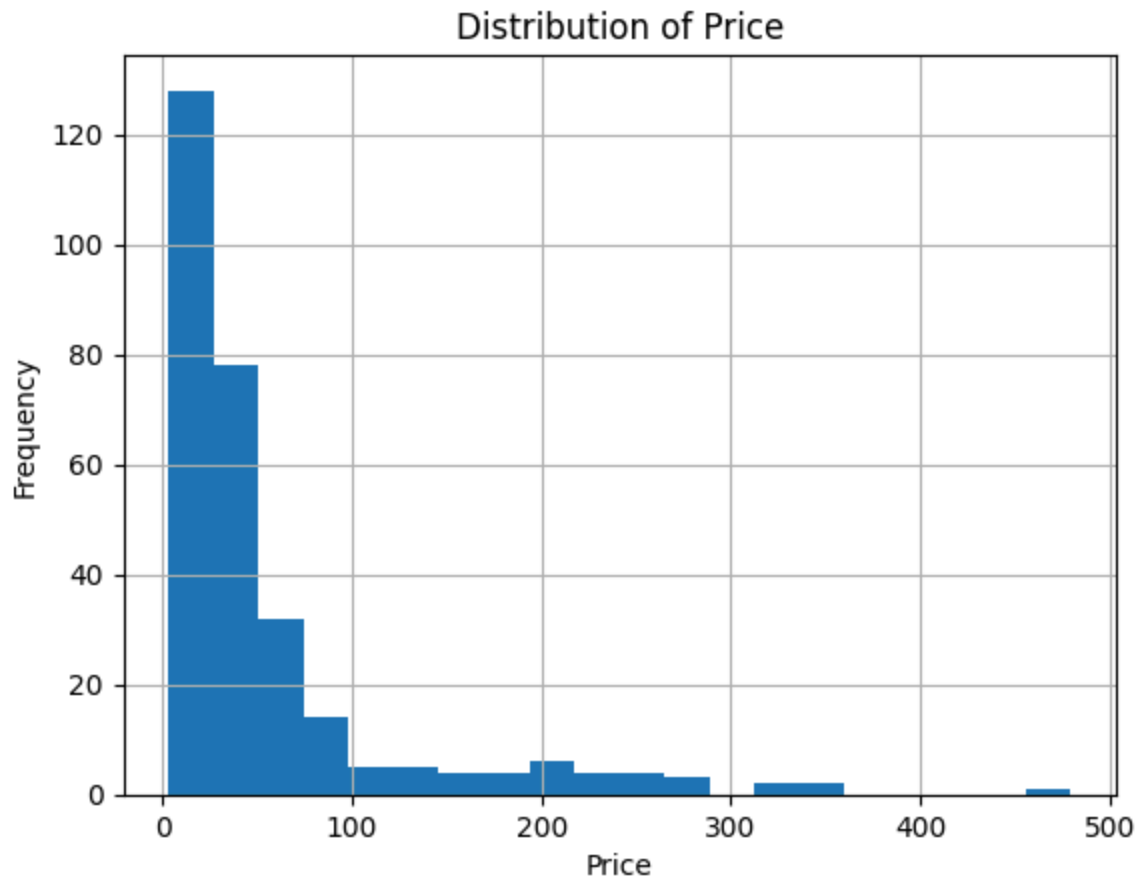
```
In [2]: for col in ["price", "qty"]:
    print(col, "mean:", round(df[col].mean(), 2),
          "median:", round(df[col].median(), 2),
          "std:", round(df[col].std(), 2))
```

price mean: 55.52 median: 37.53 std: 71.77

qty mean: 2.99 median: 3.0 std: 1.41

I would report the median (37.53) for price. The mean (55.52) is skewed by high-priced items and the median is more resistant to being skewed.

```
In [3]: df["price"].hist(bins=20)
plt.xlabel("Price")
plt.ylabel("Frequency")
plt.title("Distribution of Price")
plt.show()
```



The histogram is right skewed.

```
In [4]: df["revenue"] = df["price"] * df["qty"]
df.head()
```

```
Out[4]:
```

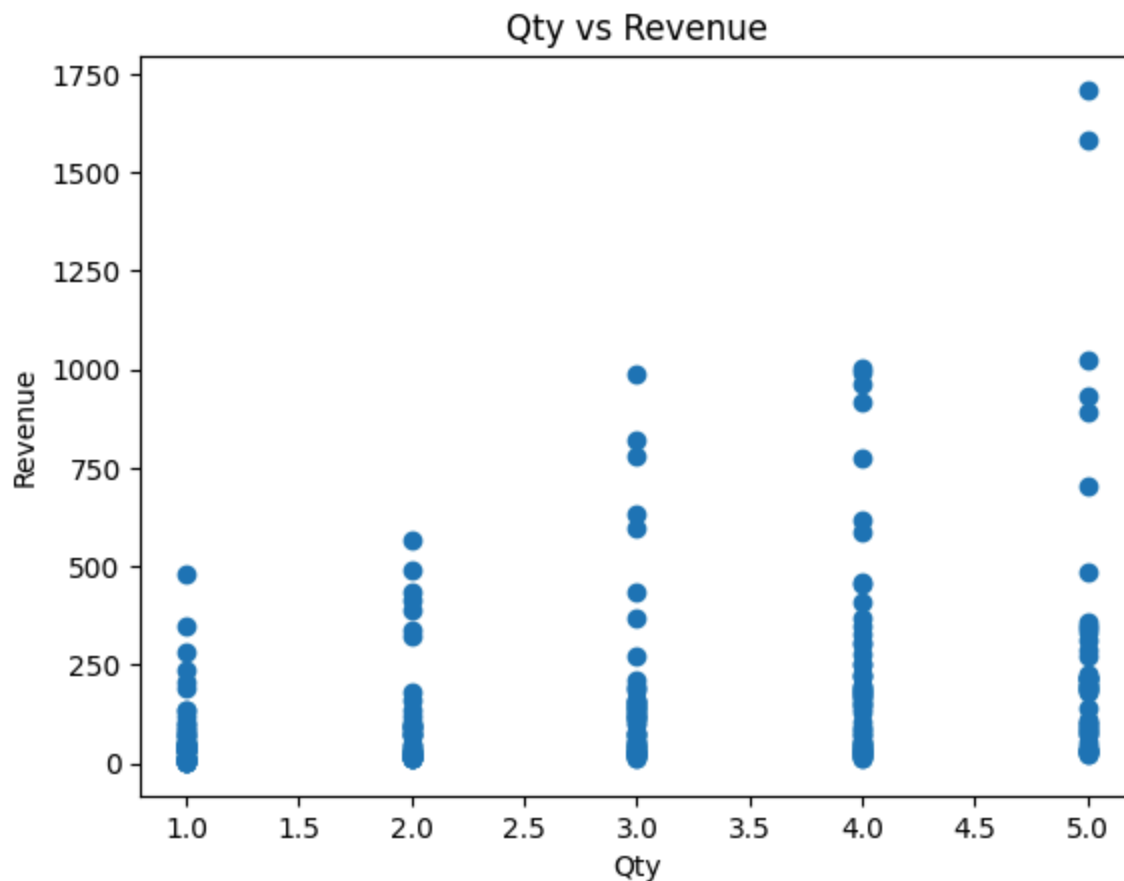
	order_id	date	product	price	qty	zip	revenue
0	1254	2026-04-06	mug	11.36	4	60614	45.44
1	1057	2026-05-30	charger	14.79	3	30303	44.37
2	1150	2026-05-06	mug	5.50	1	10001	5.50
3	1066	2026-05-30	notebook	37.53	1	98101	37.53
4	1114	2026-04-06	webcam	45.59	4	90405	182.36

```
In [5]: df[["price", "qty", "revenue"]].corr()
```

```
Out[5]:
```

	price	qty	revenue
price	1.000000	-0.041982	0.821001
qty	-0.041982	1.000000	0.312793
revenue	0.821001	0.312793	1.000000

```
In [6]: plt.scatter(df["qty"], df["revenue"])
plt.xlabel("Qty")
plt.ylabel("Revenue")
plt.title("Qty vs Revenue")
plt.show()
```



Revenue is driven by price ($r \approx 0.82$) more than by quantity ($r \approx 0.31$). This correlation doesn't prove causation, and a few very expensive outliers could be swaying the results.

sorted data is 2, 4, 4, 6, 9. Mean = $(2+4+4+6+9)/5 = 25/5 = 5.0$ Median (middle of 5 sorted values) = 4

```
In [7]: import numpy as np
data = [2, 4, 4, 6, 9]
print(np.mean(data), np.median(data))
```

5.0 4.0